CHARACTERIZATIONS OF QUASI-SEPARATED ALGEBRAIC SPACES BY QUASI-COMPACTNESS OF FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 04SG: sources/stacks-chow/tags.txt, source lines 6219–6219. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03KD: sources/stacks-chow/tags.txt, source lines 4673–4673. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JS: sources/stacks-chow/tags.txt, source lines 2025–2025. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01K4: sources/stacks-chow/tags.txt, source lines 2037–2037. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent*

1. *X is a quasi-separated algebraic space,*
2. *for $U \to X$, $V \to X$ with U, V quasi-compact schemes the fibre product $U \times_X V$ is quasi-compact,*
3. *for $U \to X$, $V \to X$ with U, V affine the fibre product $U \times_X V$ is quasi-compact.*

Proof. Using Spaces, Lemma 04SG we see that we may assume $S = \mathrm{Spec}(\mathbf{Z})$. Since $U \times_X V = X \times_{X \times X} (U \times V)$ and since $U \times V$ is quasi-compact if U and V are so, we see that (1) implies (2). It is clear that (2) implies (3). Assume (3). Choose a scheme W and a surjective étale morphism $W \to X$. Then $W \times W \to X \times X$ is surjective étale. Hence it suffices to show that

$$j : W \times_X W = X \times_{(X \times X)} (W \times W) \to W \times W$$

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

is quasi-compact, see Spaces, Lemma 03KD. If $U \subset W$ and $V \subset W$ are affine opens, then $j^{-1}(U \times V) = U \times_X V$ is quasi-compact by assumption. Since the affine opens $U \times V$ form an affine open covering of $W \times W$ (Schemes, Lemma 01JS) we conclude by Schemes, Lemma 01K4. $\square$